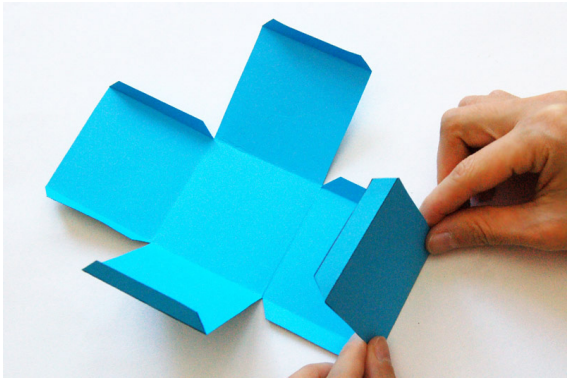
Lab: UV Mapping



Open up *table.blend* file. Inside you'll see a 3D table. Set the preview to Viewport Shader and you'll a wood texture being applied. The wood texture looks good on the top of the table, but if we look at the legs of the table, the texture appears stretched. Why is that? We'll allow the 3D mesh makes sense to us, the program Blender is not 100% sure how to apply a flat object such as the wood texture to a 3D object like the table.

In this table we'll learn about UV editing. UV editing lays out the 3D model into a 2D shape, which will make it easier to apply and paint onto our object.

Setup

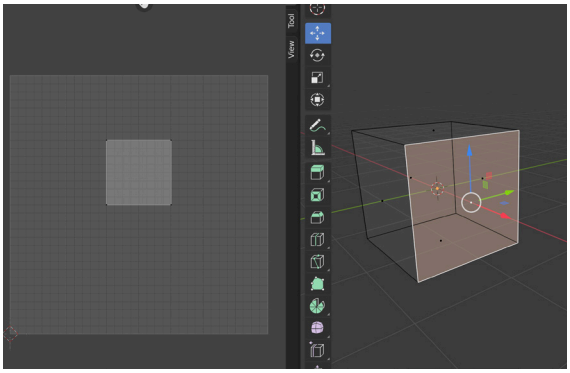


Begin by opening up *uv-editing-1.blend*. Inside you'll see a cube.

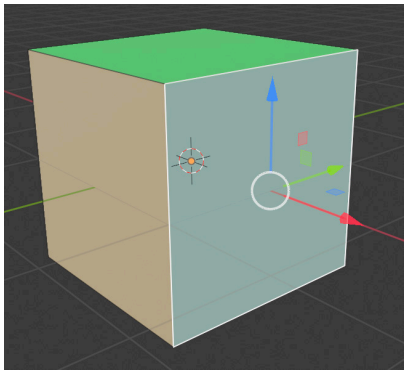
Before we begin it is important to understand what UV mapping is. Have you ever built a 3D object out of paper or cardboard? You may have cut a piece of paper into a t-shape and then applied folds to create the shape. This could also apply to assembling a pizza box or take-out box. UV mapping is a similar idea of how would we build our 3D mesh in Blender out of folded object.

Begin by clicking on the UV Editing tab in Blender. This will setup Blender to show your 3D Viewport on the right and what is called the UV Editor on the left.

Select Faces



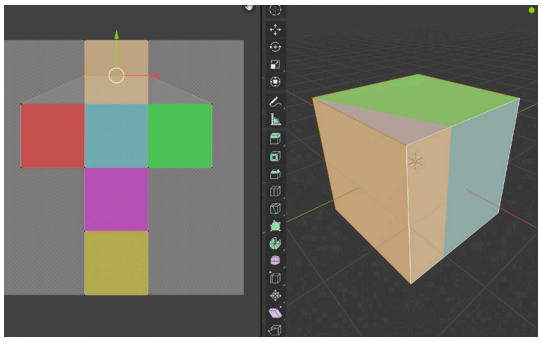
Begin by putting the cube into *Edit Mode* and set the front face of the cube. You'll notice that a white line appears in the UV Editor. That white line is the UV for that part of the cube. Switch to the wireframe mode and select all the faces. You'll notice that a t-shape appears in the UV Editor. This shape is how Blender will layout a texture.



Highlight the cube and apply the Base Material to it. Set your scene to Materials Preview to view the multicolored texture being applied as a material, you'll also see that the image has been applied to the background of the UV editor.

Highlight the light blue mesh face on the front of the cube. You'll see that it corresponds to the blue part of our image in the UV Editor. Highlight the green mesh face and you'll see that it corresponds to the green part of the image, etc. In other words, our mesh is being mapped to parts of our 2D image. If you highlight all of the faces in the cube you'll see that the correspond to a part of the 2D image.

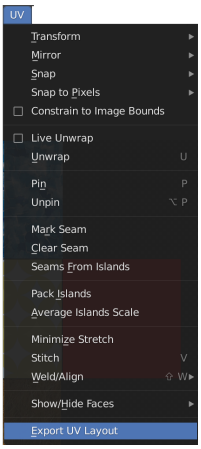
Edit the UV Map



Now that we have a basic understand of what a UV map does, let's play around with it! Begin by selecting the front face of the cube. If you go the UV Editor you'll see that it gives you the corresponding vertices. We can select those vertices in the UV Editor and edit them by moving, scaling and rotating them, just like 3D meshes. I took the two vertices at the top and moved them up. You'll see that square now overlaps the blue and yellow of the background image. If I go back and view the cube in the Shader Viewport I can see that the texturing of my cube has changed.

Go and edit the verticies yourself and view how they change the texture of your cube.

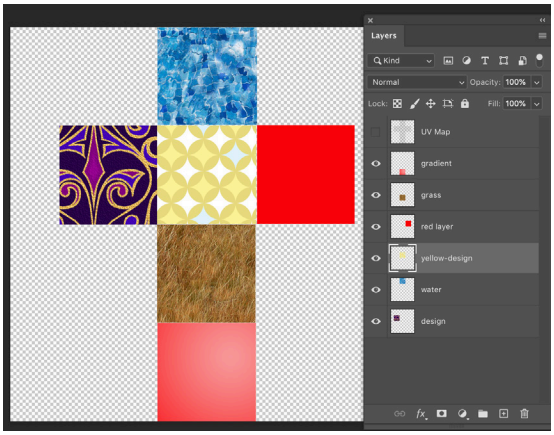
Exporting UVs



Now that we have a basics understand of what UVs are and what they do in terms of 3D meshes, let's learn about exporting uvs. Why would we want to export UVs? Well, exporting the UVs of an object makes it easy to bring those UVs into an image editing program like Adobe Photoshop or similar program. We can then use those uvs as reference when we draw into them.

Begin selecting the all the faces in the cube. The easiest way to do this to put the cube in wireframe and select all of the faces.

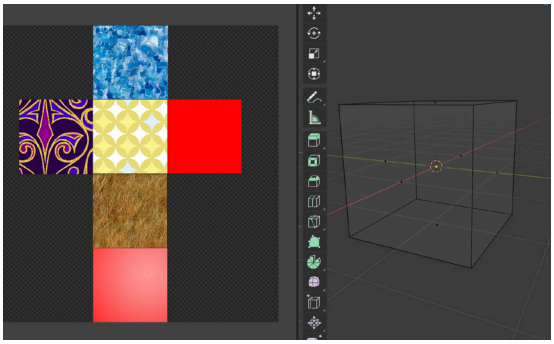
Create UV Texture Map



In the UV Editor go to *UV > Export UV Layout* and save the image to your computer. This image contains the UV maps of your cube. Take the UV image into an image editing program like Adobe Photoshop or GIMP, I will be using Adobe Photoshop for this lab.

In your image editing program put the UV image at the top of your layers as a reference. In my example I locked the layer. Then proceed to use the different drawing tools to fill in each face of the UV with a different color, image, gradient or pattern. Save your UVs as a PSD file, in case you want to go back and make changes to the layers and the file. Turn off the UV layer so it will not appear in your texture image itself.

Import Texture Map



Now that we created a new texture map based on the cube UVs go back to Blender. Create a new material called 'New Cube Material' and apply an image texture. Apply your new texture image and you should see it map out. Viola! Now you are able to apply imagery to a very specific part of a 3D mesh.

Embed Texture File

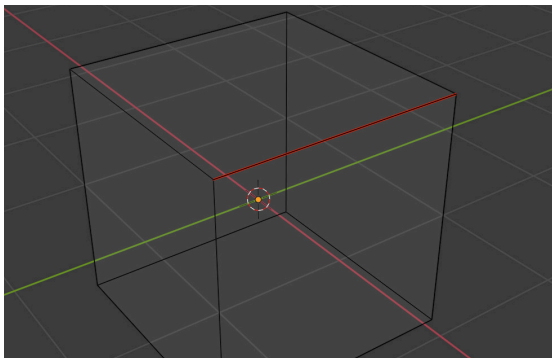
Now that we've created our own texture file let's embed it into the .blend file.

Go to *File > External Data > Pack All Into .blend*

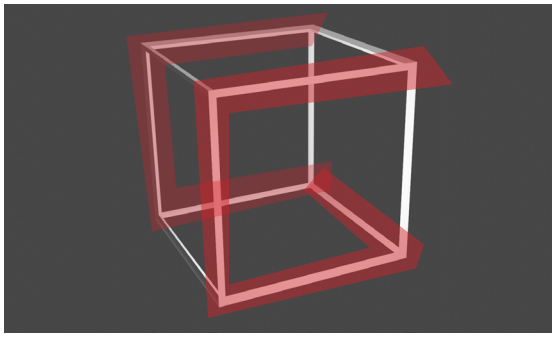
Unwrap a 3D Mesh

Some basics objects in Blender come with a UV map like a cube or sphere, but sometimes we'll need to create our own UV map. Creating a UV map is the process of taking the 3D mesh and thinking about how to cut the mesh to unfold into a 2D plane.

Let's use the cube on the right, if we make cuts to the side we can unfold or separate the faces of the cube into what is displayed. We'll do the same thing, but digitally in Blender.



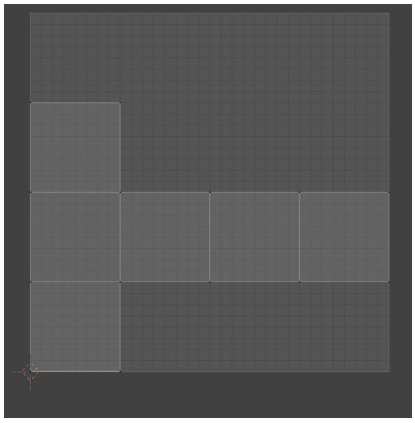
Let's begin by open up *uv-editing2.blend*. In this example we'll unwrap the cube in the scene. Begin by putting the cube into *Edit Mode*. We'll start by selecting an edge and choose *UV > Mark Seam*, this will make the edge a reddish color. A seem will eventually be a cut for our 3D mesh. To remove a seem you made select the edge and go to *UV > Clear Seam* and it will disappear.



Try to select the seams as I have in the diagram.

After you set your seams it is time to unwrap your 3D mesh into a UV map. To unwrap your shape select all of the faces of the 3D mesh in Edit Mode and go to *UV > Unwrap*. You should see the UV map appear in the UV editor.

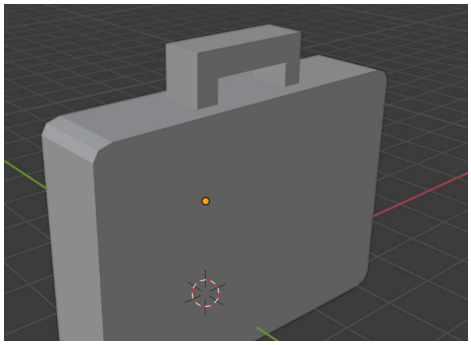
If you do not like your UV map you can go back to the mode and mark / clear more seams. After changing the seams you'll need to select the object again and go to *UV > Unwrap* again to see the changes.



Your unwrapped UV map should look like the one the right.

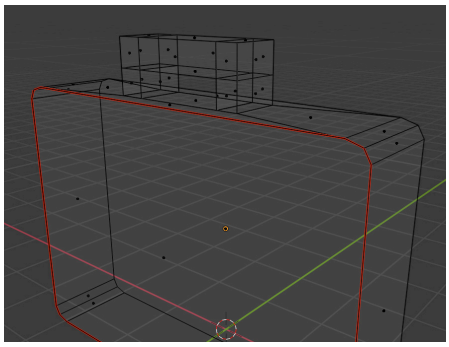
Unwrap a Complex Object

Now that we understand about UV maps and unwrapping a 3D mesh let put it all together to unwrap and create a texture for a complex object.



Setup

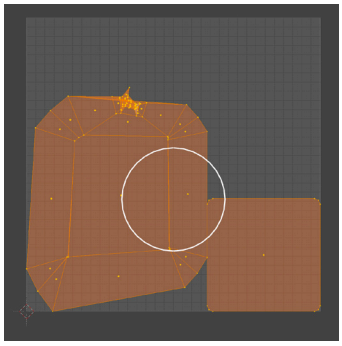
Open *uv-editing3.blend* and you'll see a old-fashioned 70's style suitcase mesh. We want it to look the image on the right that has a leather texture, complete with travel stickers!



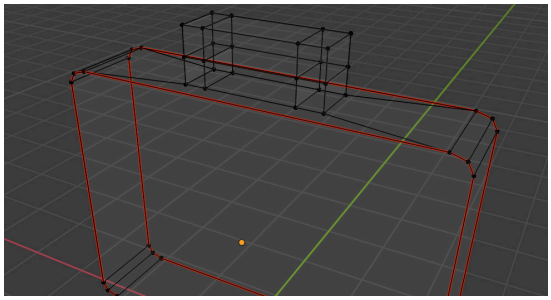
Let's begin first by creating a UV map and then we'll export that map to an image editor like Adobe Photoshop.

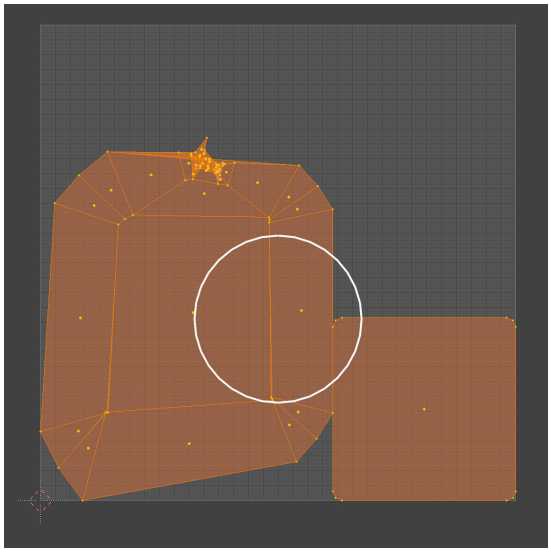
UV Map

If you look at the suitcase, the main areas are the front and back of the suitcase, so let's start with that.



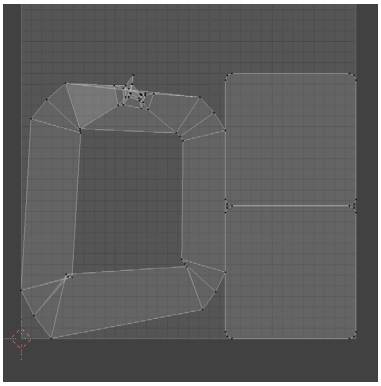
Put the suitcase in Edit Mode and select all of the edges of the front part of the suitcase. Mark these as seams by going to *UV > Mark Seam*. If you select all of the mesh and unwrap it by going to *UV > Unwrap* it will look the image on the right. You can see we are getting the front of the suitcase.





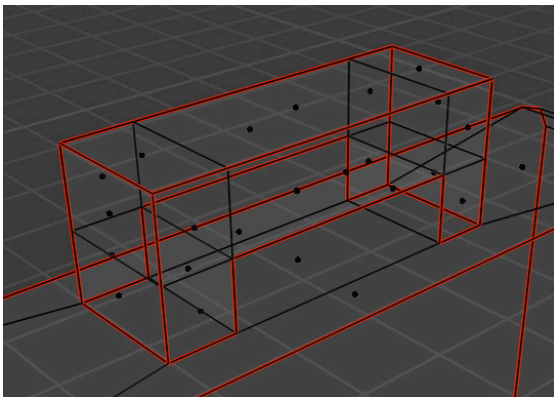
Continue Unwrapping

Let's continue to unwrap the suitcase. Select all of the edges for the back of the suitcase and mark them as seams. If you select the entire mesh and unwrap it by going to *UV > Unwrap* it will look like the image on the right.



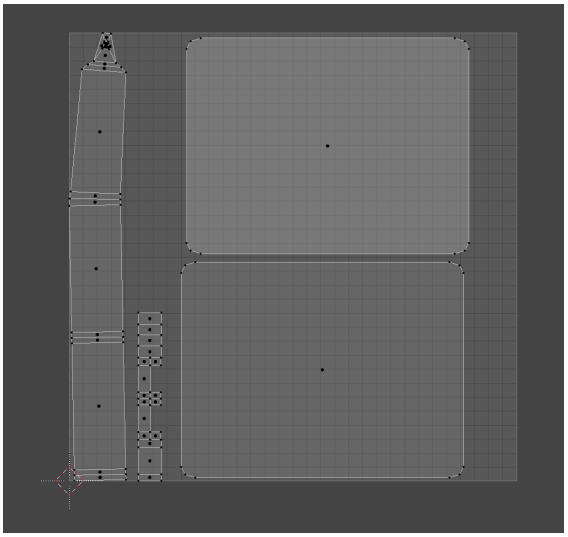
The Rest

We have the most important areas unwrapped, but we need to work on the trim of the suitcase and the handle. The trim is really a loop that goes around the suitcase. So, if we were to make a single cut we could unwrap it. Add a seam like the one in the image on the right and unwrap your model again.

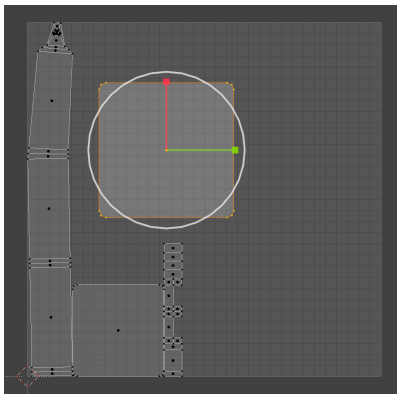


Finally, let's remove the handle of the suitcase. The handle should be separeted because we are going to apply a different texture to it.I recommend separate the handle into sides; the left side, right side, top and bottom as shown in the diagram on the right.

Select and unwrap again your UV map should look roughly like the one on the right.



Edit the UV Map



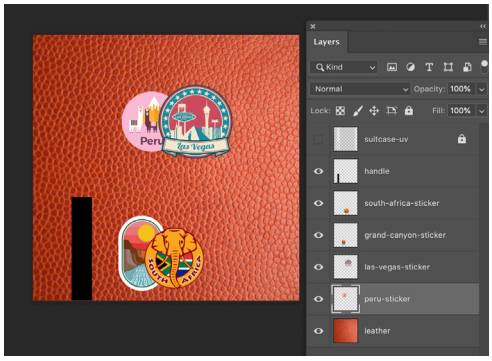
Now that we've unwrapped our suitcase we may want to edit the UVs of the map. The front and back of our suitcase are really the most important parts of the model, therefore we should make them the largest. We can do this by highlighting the front 3D mesh face. You'll see it in the UV map and we can use the move / scale tool to make it larger. Try to make the front and back face large and fit them into the UV map.

Export the UV / Create Texture Map

Highlight your UV map in the UV Editor and go to *UV > Export UV* Layout. Open the exported UV Map into an image editing program such as Adobe Photoshop / GIMP.

Open the image *leather.jpg* into your image editor.This texture will be the basis of our suitcase. Copy it into your UV map file.

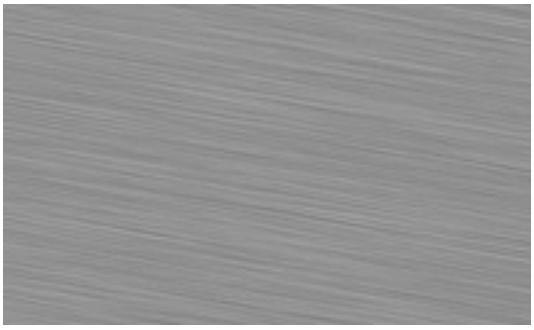
Stylize the Suitcase!



One of the benefits of creating our own UV map is that we can place textures and images exactly where we want them. Now that we have a base leather for our suitcase, let's add some fun travel stickers!

Inside the texture folder for the lab go to stickers. Choose a sticker and open it into your image editor. Place it within the texture file. Use the various photo editing tools to move, scale and rotate the sticker into the position that you want. Repeat this with at least two more stickers.

More Textures



Now that we have the main part of the suitcase textured, let's add a texture for the handle. For this lab we are going to make the handle of the suitcase a silver aluminum texture. Open the aluminum.jpg texture inside of the Textures folder and bring it into your UV-Texture map. Use it to texture the handle.

Export and Apply



Now that we've created a detailed texture map, save it as an uncompressed file such as a .psd, .tiff or .pxd. Now let's export a flattened file for Blender. Save your file as *suitcase-texture.png*.

Go back to Blender and create a material for your suitcase called suitcase material. For the base color apply a texture image and choose the *suitcase-texture.png*. Put the model into Material Preview mode and it should look something like the image on the right.

Last Bits and Submit

Now that our 3D mesh is textured, let's make sure that the our texture file in embedded inside the .blend file. In Blender go to *File > External Data > Pack All Into .blend*. This will embed your suitcase texture file into the Blender file.

Submit a folder that contains all three Blender files.